



A quantum group consensus method incorporating fairness theory[☆]

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ABSTRACT

In the complex and uncertain group decision-making environment, there is interference between decision makers, and their psychological perception has an important impact on decision-making behavior, and then affects the reaching of group consensus. To address this issue, a multi-attribute quantum group consensus model is proposed. Firstly, based on the normal cloud theory, an optimization model is constructed to determine the optimal weight of each attribute. Secondly, combining the contribution level and psychological perception of decision makers, a group consensus optimization model is constructed to promote the formation of group consensus on the basis of taking into account the fairness concerns of decision makers and different risk avoidance levels. On this basis, a quantum-like Bayesian network is constructed to deal with the interference effect between decision makers, calculate the quantum probabilities of alternatives and rank them. Finally, a numerical case validates the model, and its robustness is tested through sensitivity analysis, simulation analysis, and comparative analysis. The results show that moderate risk sensitivity and confidence can improve group consensus and reduce adjustment cost.

1. Introduction

In complex and uncertain group decision-making environments, achieving consistency among diverse opinions has become a central issue in multi-attribute group decision-making (MAGDM) research (Cao et al., 2021; Wang et al., 2022). Due to differences in individual preferences, cognitive backgrounds, and external disturbances, significant divergences often arise in decision makers' evaluations of alternative options (Chen, Zhou et al., 2023). To attain a unified group opinion, group consensus mechanisms have been widely incorporated into the decision-making process to reconcile differing views and enhance the acceptability of the final decision outcomes (Dong et al., 2016; Xu et al., 2023). Existing studies have typically realized group consensus through three types of models: the minimum adjustment cost model (Guo, Wang et al., 2024), the maximum expert participation model (Ji & Ma, 2023), and the threshold-based consensus control model (Ji, Wu, Chiclana, Sun, Herrera-Viedma, 2024; Ji, Wu, Chiclana, Sun, Liang et al., 2024). Among them, the minimum cost consensus model has been extensively applied in MAGDM due to its clear structure and well-defined optimization objective. The fundamental idea of this model is to minimize the distance between individual evaluations and the

proposed feedback, thereby reducing the overall cost of information adjustment (Xing et al., 2024).

To further enhance the behavioral rationality and acceptability of group consensus mechanisms, recent studies have increasingly incorporated fairness perception into consensus modeling frameworks (Chen, Zhu, Wang, Tsang, 2023; Du et al., 2022; Jing & Chao, 2021). As a fundamental psychological construct in group behavior, fairness perception reflects individuals' subjective evaluations of the degree to which they feel respected during the decision-making process (Jiang et al., 2025; Shen, Ma, Xu et al., 2025). In MAGDM problems, the importance of fairness perception is primarily manifested in two aspects. On one hand, if individuals perceive that their opinions are ignored during consensus adjustment, or if the feedback provided deviates excessively from their original evaluations, they may develop resistance toward the final consensus—even if the adjustments are mathematically reasonable—which ultimately reduces the acceptability of the consensus outcome (Tang & Liao, 2024). On the other hand, fairness perception fosters active participation and strengthens group cohesion, thereby improving the overall efficiency of collaborative decision-making (Gong et al., 2023). Fairness perception is

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typically categorized into three dimensions: distributive fairness, procedural fairness, and interactional fairness (Colquitt, 2001). Among these, procedural fairness plays a particularly prominent role in consensus modeling, as it pertains to whether individuals perceive the adjustment process itself as fair. For instance, Guo et al. introduced an incentive-and-penalty mechanism to guide decision makers' behavior during the consensus process, thereby enhancing both the fairness and objectivity of the consensus mechanism (Guo, Zhang et al., 2024). Liu et al. (2025) developed a fairness-based utility function and integrated it with an acceptance threshold for adjustments, achieving a balance between information fidelity and fairness optimization. Liu and Zhu (2022) further proposed the incorporation of a fairness perception factor into the consensus adjustment process to increase decision makers' satisfaction and, in turn, improve the overall level of group consensus. Chen, Zhu, Wang, Chiclana et al. (2023) proposed a method for generating group opinions that integrates fairness concerns, emphasizing that introducing fairness theory in group decision-making helps to more truly depict the behavioral characteristics and psychological perceptions of individuals in the process of consensus formation. These studies collectively suggest that embedding fairness perception into consensus modeling not only improves the acceptability of feedback mechanisms but also provides theoretical support for enhancing the efficiency and quality of group consensus.

Building upon the foundation of fairness perception mechanisms, measuring and allocating individual marginal contributions has become a core element in ensuring adjustment fairness and maintaining the stability of group cooperation (Shen, Ma, Kou et al., 2025). Given that participants in group decision-making often exhibit self-interested behavior, failing to properly assess each individual's actual role in achieving consensus may lead to a lack of recognition or acceptance of adjustment strategies, thereby undermining the stable realization of consensus. In response, cooperative game theory has been widely employed in consensus optimization modeling, aiming to enhance the fairness and sustainability of consensus outcomes through marginal benefit allocation mechanisms (Meng, Tang et al., 2023). Among various solution concepts in game theory, the Shapley value has been extensively used to evaluate each decision maker's contribution to consensus formation, owing to its strong interpretability and marginal fairness properties (Meng, Gong et al., 2023). By quantifying the individual impact on collective outcomes, the Shapley value serves as a basis for guiding the allocation of adjustment costs. Furthermore, enhanced Shapley-based mechanisms that incorporate factors such as interpersonal trust and willingness to adjust have been proposed to improve the adaptability and rationality of feedback mechanisms, thereby promoting higher levels of group consensus and cooperation stability (Liao et al., 2024). Therefore, the integration of the Shapley value not only strengthens the incentive effect of feedback adjustment but also provides a solid theoretical foundation for the development of fair and efficient group consensus models.

In addition to optimizing group consensus mechanisms, the handling of uncertainty at the attribute level constitutes another critical aspect that cannot be overlooked in MAGDM. Traditional approaches to attribute weight determination are generally categorized into two types: subjective weighting methods and objective weighting methods. Subjective empowerment methods, such as analytic Hierarchy Process (AHP) and Delphi method, rely on the experience and judgment of decision makers. Although they can combine human knowledge and experience, they are too dependent on subjective opinions and susceptible to personal preferences, making it difficult to maintain objectivity and consistency in a complex and uncertain environment (Duleba & Blahota, 2022; Yang et al., 2023). On the other hand, objective weighting methods, such as entropy weight method and variance method, determine weights based on the objective characteristics of data. Although the interference of subjective factors is avoided, the randomness and fuzziness of data are often ignored, and it is difficult to deal with uncertainty in real decision making (Chen, 2021; Liu et al., 2021).

Both subjective and objective weighting methods are inadequate in the face of complex, dynamic and uncertain decision-making background, especially the lack of effective processing of data fuzziness and randomness. In recent years, cloud model, as an uncertainty theory tool that can deal with ambiguity and randomness at the same time, has shown significant advantages in solving complex decision problems. The core of the cloud model is to relate qualitative concepts and quantitative data through the mapping of uncertainty, so as to deal with practical problems involving stochastic fluctuations and fuzzy boundaries (Rao et al., 2022; Wan et al., 2024).

Following the completion of the group consensus process, how to effectively utilize and rank the agreed-upon evaluation information emerges as another key task in MAGDM research. Under complex cognitive contexts, decision makers may still exhibit latent psychological interference and cognitive coupling effects even after reaching consensus. In this regard, quantum decision theory—by introducing concepts from quantum mechanics—offers a novel theoretical framework for interpreting irrational behaviors that are difficult to explain using traditional decision-making theories (Xu, Yan, Li, 2025; Yan et al., 2025). Interference phenomena in quantum decision theory are not limited to the interaction of individuals' internal cognitive states, but can also be used to explain the mutual interference among decision makers (Busemeyer, 2012; He et al., 2018; Moreira et al., 2020; Yan, Xu et al., 2024). Specifically, when multiple DMs are involved in a decision, each decision maker's choices and preferences affect each other, similar to interference effects in quantum mechanics (Xu, Yan, Pérez et al., 2025). The cognitive and emotional states of different decision makers overlap and interfere with each other in the group decision-making process, resulting in changes in the overall decision-making results (Jiang & Liu, 2022; She et al., 2021; Wu et al., 2022). Through the perspective of quantum interference, the interaction between decision makers is no longer independent, but interrelated, thus changing the final group decision outcome. For example, Tan et al. (2022) combined foreground theory and quantum probability theory to establish a consensus improvement mechanism for linguistic distributed group decision making. Wang, Liu et al. (2024) uses quantum probability theory to aggregate trust relationships and proposes a dual mechanism including trust exploration and leadership incubation, which is used to model opinion dynamics in social networks and promote consensus in group decision-making. Yan, Zeng et al. (2024) proposed a disturbance effect model based on minimum constraint and information entropy to measure the disturbance value in the quantum decision model, aiming at the disturbance effect problem in multi-attribute group decision making.

Despite the significant progress achieved in MAGDM, particularly in the area of group consensus modeling, several critical challenges and research gaps remain under complex and uncertain decision-making environments:

(1) Insufficient integration of decision makers' contributions and fairness perception in consensus modeling. Most existing minimum cost consensus models focus on minimizing the informational deviation between individual evaluations and feedback suggestions, often overlooking the marginal contributions of decision makers and their subjective perception of fairness in the consensus formation process. If the feedback adjustment fails to reflect an individual's relative contribution or violates their fairness expectations, it may induce cognitive resistance—even when the adjustment cost is low—thereby undermining the acceptability of consensus outcomes and the stability of group cooperation.

(2) Limited application of quantum probability theory in the ranking phase of MAGDM. In most current studies, the post-consensus ranking is conducted using linear aggregation methods, which typically assume cognitive independence among decision makers. Such methods fail to accurately capture the mutual interference behaviors and nonlinear cognitive deviations that often occur in real-world decision

environments. In practice, evaluations and feedback are interdependent, reflecting evident non-independence and cognitive entanglement. Quantum probability theory, by modeling non-classical uncertainty through interference terms, provides a powerful theoretical tool to represent such cognitive coupling. However, the literature has rarely addressed the incorporation of quantum interference structures at the group level for ranking modeling, and there is a notable lack of research on constructing quasi-quantum Bayesian structures—based on feedback information—to compute the probabilities of alternatives.

(3) Inadequate treatment of information uncertainty in attribute weight optimization. Traditional weight determination methods mainly rely on subjective preferences or objective data characteristics, making it difficult to simultaneously model the fuzziness and volatility inherent in evaluation information. In complex and uncertain decision-making scenarios, attribute assessments are often influenced by multi-source disturbances and heterogeneous cognition, exhibiting characteristics of intertwined randomness and vagueness. Although several studies have attempted to use entropy, grey relational analysis, and other techniques to capture information distribution patterns, a unified structural approach that models both uncertainty and variability is still lacking. The normal cloud model, as a powerful tool for uncertainty representation, employs a triplet structure that effectively characterizes the centrality, dispersion, and fuzzy boundaries of attribute evaluations, offering high adaptability. However, its application in existing MAGDM research remains insufficient and underexplored.

To address these research gaps, this paper proposes a multi-attribute quantum group consensus model based on cooperative game theory. Firstly, this paper introduces a risk preference function and combines superentropy in the normal cloud model to construct an optimization model for solving attribute weights. Secondly, considering the change in decision makers' psychological preferences under uncertain dynamic decision-making environments, along with their perception of fairness, a group consensus model incorporating decision makers' marginal contributions is constructed to generate the feedback and suggestion matrix. Finally, a quantum-like Bayesian network is constructed to calculate the quantum probabilities of each alternative and rank them accordingly.

The innovative contribution of this paper is mainly reflected in the following aspects:

(1) This paper constructs a group consensus model that integrates decision makers' contribution and fairness perception, measures decision makers' contribution based on Shapley values, and combines the influence of fairness perception to optimize decision makers' adjustment behavior, so as to improve the efficiency of consensus reaching in the group decision-making process.

(2) Under the background of multi-attribute decision making, this paper proposes a multi-attribute quantum group decision making model combined with quantum decision theory, analyzes the interaction of quantum interference effect among multiple decision makers, and explores the mechanism of the effect on group consensus formation.

(3) This paper proposes a weight optimization method based on normal cloud model, which uses entropy and overentropy to describe uncertainty and volatility in attribute evaluation, respectively, and optimizes the weight calculation process in multi-attribute decision-making, so as to better deal with fuzziness and uncertainty in decision-making.

The structure of this paper is as follows: The Section 2 introduces the quantum probability theory and the relevant theories of the normal cloud model, which serve as the theoretical basis of the following model. Section 3 mainly describes the structural framework of the paper. In Section 4, we introduce the multi-attribute quantum group consensus model which takes into account decision maker psychology and cooperative relationship. The Section 5 verifies the validity of the model according to a numerical case, and verifies the robustness and applicability of the model through sensitivity analysis, simulation analysis and comparative analysis. The Section 6 is to summarize the article and shortcomings, and put forward the future research direction.

2. Preliminaries

2.1. Normal cloud model

Cloud model theory (Li et al., 2009) is a theory proposed on the basis of probability theory and fuzzy mathematics to reflect the fuzziness and randomness of concepts in the human knowledge system and to study the conversion between uncertain information and quantitative indicators.

The digital characteristics of the cloud are represented by three parameters (Ex, En, He) , where the expected value Ex represents the expected value of the cloud droplet in the spatial distribution of the discourse domain.

$$Ex = \frac{1}{n} \sum_{i=1}^n x_i \quad (1)$$

where $x_i (i = 1, 2, \dots, n)$ represents a cloud droplet in the discourse domain.

The entropy of cloud En represents the uncertainty measure of the qualitative concept, reflects the dispersion degree of cloud droplets in the domain of theory, and is used to describe the span of cloud.

$$En = \sqrt{\frac{\pi}{2}} \times \frac{1}{n} \sum_{i=1}^n |x_i - Ex| \quad (2)$$

The hyper entropy He of cloud represents the uncertainty measure of entropy, which reflects the randomness of the dispersion degree and certainty degree of cloud droplets, and is used to describe the thickness of cloud.

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - Ex)^2 \quad (3)$$

$$He = \sqrt{|S^2 - (En)^2|} \quad (4)$$

2.2. Fairness preference theory

As an important branch of behavioral economics, the theory of fairness preference studies that individuals not only care about their own material benefits, but also care about the benefits obtained by others in the decision-making process, thus reflecting the preference for fairness and unfairness (Bruyn & Bolton, 2008). Unlike traditional economic theory, which assumes that individuals are perfectly rational and only seek to maximize their own interests, fair preference theory suggests that people consider a sense of fairness when allocating resources and may sacrifice some of their personal interests in order to reduce inequality.

Definition 1 (Fehr & Schmidt, 1999). The F-S model of fairness preference theory, proposed by Fehr and Schmidt in 1999, is a model that formalizes how individuals balance their own interests with social fairness.

$$U_i(x) = y_i - \frac{\alpha_i}{s-1} \sum_{j \neq i} \max(y_j - y_i, 0) - \frac{\chi_i}{s-1} \sum_{j \neq i} \max(y_i - y_j, 0) \quad (5)$$

where, $U_i(x)$ represents the individual's utility, $y_i (i = 1, 2, \dots, s)$ and $y_j (j = 1, 2, \dots, s)$ represent the absolute income utility of decision makers $d_i (i = 1, 2, \dots, s)$ and $d_j (j = 1, 2, \dots, s)$, respectively. $\alpha_i (0 \leq \alpha_i \leq 1)$ represents jealousy, and $\chi_i (-1 < \chi_i < 1)$ represents the coefficient of sympathy.

2.3. Quantum probability theory

Definition 2 (Yan et al., 2025). A Hilbert space is a complete complex vector space equipped with an inner product, denoted as $\mathcal{H}$. In this space, each vector $|\psi\rangle \in \mathcal{H}$ and $|\phi\rangle \in \mathcal{H}$ represents a quantum state of the system. If the inner product between two vectors is zero,

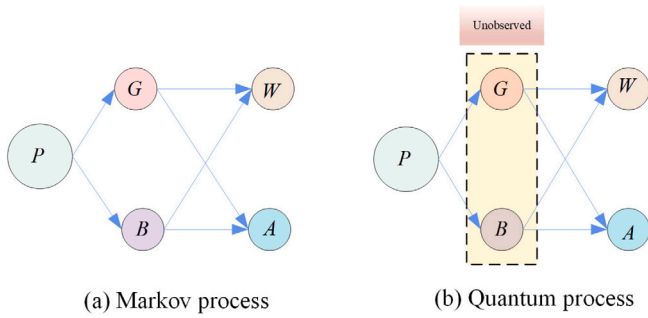


Fig. 1. Decision making in Markov and quantum processes.

that is $\langle \psi | \phi \rangle = 0$, the vectors are defined as orthogonal. The inner product in Hilbert space defines a measurement mechanism, where the quantum system collapses into an eigenstate during measurement, and the probability of an outcome is as follows:

$$P = |\langle \phi | \psi \rangle|^2 \quad (6)$$

where $|\psi\rangle$ is the system's initial state, $|\phi\rangle$ is the eigenstate corresponding to the measurement, and $\langle \phi | \psi \rangle$ represents their inner product.

Quantum Probability Theory (QPT) builds a probabilistic framework for describing state superposition, interference, and measurement uncertainty, based on the structure of Hilbert space (Bussemeyer et al., 2024). Unlike classical probability, which assumes the independence of events, QPT introduces quantum superposition and interference effects. As shown in Fig. 1(a), classical Markov processes follow a linear and independent path structure, where events do not interfere with one another. In contrast, as illustrated in Fig. 1(b), in quantum processes, the system can exist in a superposition of different paths, and these paths are correlated by phase relationships, leading to non-linear and non-classical decision behaviors.

According to Feynman's first rule of path integrals, if a system evolves through multiple paths to a final state, the probability amplitude of each path is the product of the amplitudes of each segment along the path (Xu, Yan, Cabrerizo, 2025). The total amplitude of the system is the sum of the amplitudes of all paths, and the total probability is the square of the total amplitude. Based on this principle, if the system may reach the final state W via two intermediate states G and B , the probability of W is as follows:

$$\begin{aligned} P(W) &= |\psi_G \psi_{W|G} + \psi_B \psi_{W|B}|^2 \\ &= |\varphi_G e^{i\theta_G} \varphi_{W|G} e^{i\theta_{W|G}} + \varphi_B e^{i\theta_B} \varphi_{W|B} e^{i\theta_{W|B}}|^2 \\ &= |\varphi_G e^{i\theta_G} \varphi_{W|G} e^{i\theta_{W|G}} + \varphi_B e^{i\theta_B} \varphi_{W|B} e^{i\theta_{W|B}}|^2 \\ &= P(G) P(W|G) + P(B) P(W|B) + \\ &\quad 2\sqrt{P(G) P(W|G) P(B) P(W|B)} \cos(\theta_G - \theta_B) \end{aligned} \quad (7)$$

where, $\psi_G = \varphi_G e^{i\theta_G}$ and $\psi_{W|G} = \varphi_{W|G} e^{i\theta_{W|G}}$ represent the complex probability amplitudes corresponding to the path G and its transition to W .

2.4. Group consensus basic model

In MAGDM problems, due to differences among decision-makers in terms of knowledge structure, experiential background, and risk preferences, their original preferences often exhibit significant divergence (Cao et al., 2024). Such inconsistencies can hinder the achievement of group consensus and negatively affect the coordination and implementation of the final decision (Zhang et al., 2023). To address this issue while preserving the authenticity of individual preferences,

the Minimum adjustment consensus model has been proposed. The model aims to adjust individual preferences with the minimal possible deviation, under the constraint that the adjusted preferences must satisfy a predefined consensus threshold. The optimization model (Zhang et al., 2020) can be formulated as follows:

$$\begin{aligned} \text{Min} \quad & \sum_{k=1}^s d(z^k, \hat{z}^k) \\ \text{s.t.} \quad & \begin{cases} \hat{z}^o = F_\pi(\hat{z}^1, \hat{z}^2, \dots, \hat{z}^s) \\ CD(z^k, \hat{z}^o) \leq CL, \forall k = 1, \dots, s \end{cases} \end{aligned} \quad (8)$$

where z^k denotes the original preference of the k th decision-maker, $\hat{z}^k$ represents the adjusted preference, and $\hat{z}^o$ is the collective preference derived through an aggregation function $F_\pi(\cdot)$. The function $d(z^k, \hat{z}^k)$ measures the distance between the original and adjusted preferences, while $CD(z^k, \hat{z}^o)$ quantifies the consensus degree between an individual preference and the collective preference. The parameter $CL \geq 0$ specifies the acceptable threshold for consensus.

3. Problem description and resolution framework

3.1. Descriptions of the multi-attribute group decision-making problem

In MAGDM problems, decision makers are required to comprehensively evaluate multiple alternatives across multiple attributes and reach consensus at the group level through interaction and adjustment. To facilitate the construction of the model and the derivation of formulas, The basic symbols are defined as follows: the set of alternatives is denoted as $A = \{a_1, a_2, \dots, a_m\}$, where a_i ($i = 1, 2, \dots, m$) represents the i -th alternative. The set of decision makers is denoted as $D = \{d_1, d_2, \dots, d_s\}$, where d_k ($k = 1, 2, \dots, s$) represents the k -th decision maker. The weight of the decision maker is expressed as λ_k ($k = 1, 2, \dots, s$). The set of attributes is denoted as $C = \{c_1, c_2, \dots, c_n\}$, where c_j ($j = 1, 2, \dots, n$) represents the j -th attribute. The attribute's weights set of decision maker d_k ($k = 1, 2, \dots, s$) is denoted as $\omega^k = \{\omega_1^k, \omega_2^k, \dots, \omega_n^k\}$, where ω_j^k ($j = 1, 2, \dots, n, k = 1, 2, \dots, s$) represents the weight of attribute c_j ($j = 1, 2, \dots, n$) assigned by decision maker d_k ($k = 1, 2, \dots, s$). The initial evaluation information matrix of decision maker d_k ($k = 1, 2, \dots, s$) is denoted as $Z^{(k)} = (z_{ij}^{(k)})_{m \times n}$, where $z_{ij}^{(k)}$ ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$) indicates the initial evaluation value given by decision maker d_k ($k = 1, 2, \dots, s$) for alternative a_i ($i = 1, 2, \dots, m$) under attribute c_j ($j = 1, 2, \dots, n$).

3.2. Collection and processing of decision information

Firstly, the evaluation information of each decision maker is collected and the initial evaluation information matrix $Z^{(k)}$ ($k = 1, 2, \dots, s$) of each decision maker is obtained:

$$Z^{(k)} = \begin{pmatrix} z_{11}^{(k)} & z_{12}^{(k)} & \dots & z_{1n}^{(k)} \\ z_{21}^{(k)} & z_{22}^{(k)} & \dots & z_{2n}^{(k)} \\ \vdots & \vdots & \ddots & \vdots \\ z_{m1}^{(k)} & z_{m2}^{(k)} & \dots & z_{mn}^{(k)} \end{pmatrix}, k = 1, 2, \dots, s \quad (9)$$

In multi-attribute decision-making problems, attributes often have different dimensions and value ranges, so directly using raw data for analysis can lead to weight imbalances between attributes, which may compromise the fairness and accuracy of the decision results. Therefore, normalizing the evaluation information matrix is necessary to eliminate dimensional differences between attributes by converting data into a unified scale.

$$z_{ij}^k = \begin{cases} \frac{z_{ij}^k - \min_i z_{ij}^k}{\max_i z_{ij}^k - \min_i z_{ij}^k}, & \text{if } c_j \text{ is a benefit attribute} \\ \frac{\max_i z_{ij}^k - z_{ij}^k}{\max_i z_{ij}^k - \min_i z_{ij}^k}, & \text{if } c_j \text{ is a cost attribute} \end{cases} \quad (10)$$

$i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$

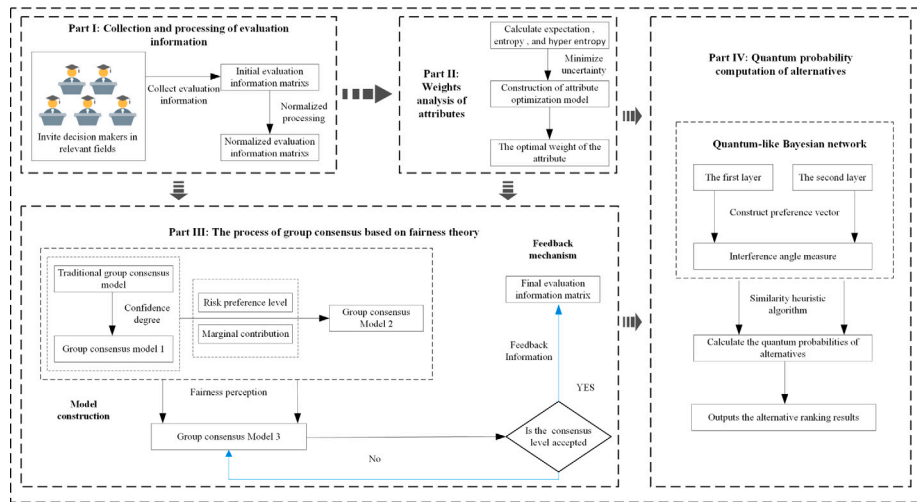


Fig. 2. The resolution framework.

3.3. The resolution framework

The framework of the method proposed in this paper is shown in Fig. 2, which mainly includes the following four processes:

(1) Collection and processing of evaluation information. In this process, the initial evaluation information provided by each decision maker for different alternatives and attributes is collected. Based on this information, normalization is performed to ensure that the evaluation data for different attributes are comparable, avoiding the impact of dimensional differences on subsequent calculations.

(2) Determination of attributes' weights. In this stage, an attribute weight optimization model is constructed based on the normal cloud model. By calculating the expected value, entropy, and hyper-entropy of each attribute, the uncertainty and volatility in attribute evaluations are quantified, and an optimization model is established to minimize attribute uncertainty, thereby solving for the optimal attributes' weights.

(3) Construction of group consensus model. This stage includes three levels: Firstly, a basic consensus optimization model is constructed to promote group consensus through feedback adjustment mechanisms. Secondly, a trust interval constraint is introduced, with a penalty mechanism for adjustments that exceed the trust range, ensuring that individual preferences of decision makers are respected during the adjustment process and preventing excessive modifications. Finally, the cooperative game theory is incorporated, with Shapley values used to calculate the marginal contributions of decision makers, and a risk preference function and fairness perception coefficients are introduced to establish a comprehensive group consensus optimization model.

(4) Quantum Probability Calculation. Based on the results of group consensus adjustments, a quantum-like Bayesian framework is constructed. Firstly, the prior probability is calculated based on the decision makers' weights. Secondly, conditional probabilities are calculated by combining the adjusted feedback information and attributes' weights. On this basis, a heuristic algorithm is applied to measure the interference angles between decision makers. Finally, quantum probabilities are calculated, and the alternatives are ranked.

4. Methodology

4.1. A novel risk avoidance function

Traditional weight calculation methods often ignore the uncertainty of information and the risk aversion of decision makers in the decision-making process, which play a crucial role in the real decision-making

environment. In this paper, a psychological function of decision makers' risk aversion is constructed to systematically reflect the influence of decision makers' risk aversion characteristics on group consensus by embedding the function into group consensus process. This function can dynamically adjust the perceived risk values of different attributes, assign lower utility contributions to attributes with higher risks, and maintain higher utility values for attributes with lower risks. This design reflects the real preferences of decision makers for different risk situations, effectively weakens the influence of high-risk attributes on consensus results, and thus makes group decision making more suitable for actual situations, as follows:

Definition 3. Assuming that decision makers are not completely rational, the risk avoidance function of decision makers in the context of uncertain complex decisions is defined as follows:

$$f(R_j) = \frac{1}{1 + \gamma \cdot (R_j)^\delta}, j = 1, 2, \dots, n \quad (11)$$

where, $R_j \in [0, 1]$, $R_j (j = 1, 2, \dots, n)$ represents the decision-maker's perception of risk aversion to attribute $c_j (j = 1, 2, \dots, n)$. $\gamma (\gamma \in [0, 1])$ represents the degree of risk sensitivity and avoidance of decision makers. The larger $\gamma (\gamma \in [0, 1])$ is, the more sensitive the function is to changes in $R_j (j = 1, 2, \dots, n)$, meaning the decision maker is more risk-averse. When $R_j (j = 1, 2, \dots, n)$ is small, the risk aversion effect is still strong, so even a small risk fluctuation will trigger a large risk aversion response. The smaller $\gamma (\gamma \in [0, 1])$ is, the lower the sensitivity of the function to $R_j (j = 1, 2, \dots, n)$, which means that the decision-maker has $R_j (j = 1, 2, \dots, n)$ higher tolerance for risk, that is, only when $R_j (j = 1, 2, \dots, n)$ is larger, the risk avoidance effect will be significantly enhanced. $\delta (\delta \geq 1)$ is a parameter that controls the degree of non-linearity of the risk avoidance function, which determines the decay rate of the risk avoidance function $f(R_j)$ when the risk of attribute $c_j (j = 1, 2, \dots, n)$ increases (see Fig. 3).

Theorem 1. The risk aversion function $f(R_j)$ is monotonically decreasing for any $R_j \in [0, +\infty) (j = 1, 2, \dots, n)$.

Proof. The first-order derivation of risk avoidance function $f(R_j)$ can be obtained: $f'(R_j) = \frac{-\gamma \delta (R_j)^{\delta-1}}{(1 + \gamma (R_j)^\delta)^2}$, since all parameters are positive, $f'(R_j) < 0$, that is, risk avoidance function $f(R_j)$ is monotonically decreasing for any $R_j (j = 1, 2, \dots, n)$. $\square$

This Theorem shows that as the value of risk $R_j (j = 1, 2, \dots, n)$ increases, the value of risk aversion function $f(R_j)$ decreases. This

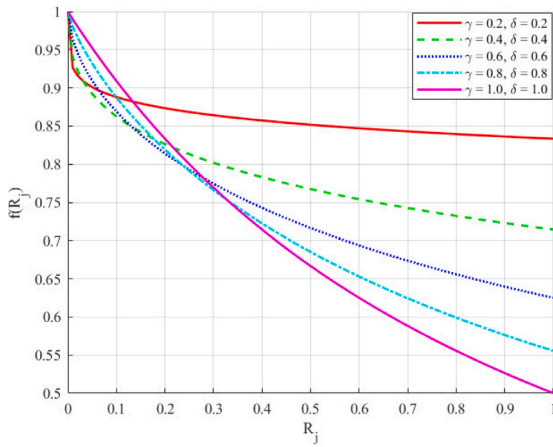


Fig. 3. A novel risk avoidance function.

is consistent with the reality that decision-makers have a stronger aversion to higher risks. In the context of multi-attribute decision making, this theorem ensures that as the risk of an attribute increases, its importance in the decision will be reduced by weight adjustment.

Theorem 2. The risk aversion function $f(R_j)$ is bounded within $R_j \in [0, +\infty)$ ($j = 1, 2, \dots, n$) and satisfies: $0 < f(R_j) < 1$.

Proof. When $R_j = 0$, then $f(R_j) = 1$, and when $R_j \rightarrow +\infty$, then $\lim_{R_j \rightarrow +\infty} f(R_j) = \lim_{R_j \rightarrow +\infty} \frac{1}{1+\gamma(R_j)^\delta} = 0$, because the risk aversion function $f(R_j)$ is monotonically decreasing for any R_j ($j = 1, 2, \dots, n$), so $0 < f(R_j) < 1$. $\square$

Theorem 3. The risk aversion function $f(R_j)$ is a convex function.

Proof. Taking the second-order derivative of the risk aversion function $f(R_j)$: $f''(R_j) = \frac{\gamma^2 \delta(\delta+1)R_j^{2\delta-2} - \gamma\delta(\delta-1)R_j^{\delta-2}}{(1+\gamma R_j^\delta)^3}$, according to the value range of the parameter, we know that $f''(R_j) \geq 0$, so the risk aversion function $f(R_j)$ is a convex function. $\square$

This theorem shows that when the risk value of an attribute increases, the increment of risk aversion will also increase. In multi-attribute decision making, this convexity ensures that high-risk attributes will be significantly penalized, which further strengthens the model's performance of high risk avoidance.

Theorem 4. If $\sum_{j=1}^n \omega_j = 1$, then for any R_j ($j = 1, 2, \dots, n$), then we can obtain $f\left(\sum_{j=1}^n \omega_j \cdot R_j\right) \geq \sum_{j=1}^n \omega_j f(R_j)$.

Proof. Since the risk aversion function $f(R_j)$ is a convex function, then $f\left(\sum_{j=1}^n \omega_j R_j\right) \geq \sum_{j=1}^n \omega_j f(R_j)$ follows from the weighted average inequality for convex functions. $\square$

This theorem states that when the risk of multiple attributes is considered, the weighted overall risk avoidance value will be greater than or equal to the weighted average of the individual risk avoidance values of each attribute. This inequality ensures that risk avoidance in the overall decision is more effective than any single attribute, helping to optimize decisions when the risks of multiple attributes are considered comprehensively.

4.2. Attributes' weights calculation based on normal cloud model

In the context of complex and uncertain decision-making, attribute evaluation information is often affected by data volatility, fuzziness, and multi-source heterogeneity, exhibiting significant randomness and uncertainty. Traditional attribute weighting methods are typically based on clear and static evaluation information, failing to fully consider the fuzziness in the evaluation process, the differences among decision makers, and the potential dependencies among attributes. As a result, they are inadequate in addressing the complexity and dynamics inherent in MAGDM. To overcome these limitations, this paper constructs an attribute weight optimization model based on the normal cloud model. By employing the triplet of expectation, entropy, and hyper-entropy in the cloud model to characterize the fuzziness and randomness of attribute values, the proposed model establishes an optimization function aimed at minimizing attribute uncertainty, thereby dynamically adjusting attribute weights. The optimization model is as follows:

$$\begin{aligned}
 M_1 : \min & \sum_{j=1}^n \omega_j^k \cdot (He)_j^k \\
 \left\{ \begin{aligned}
 \bar{z}_j^k &= \frac{1}{m} \sum_{i=1}^m z_{ij}^k & (12-1) \\
 (En)_j^k &= \sqrt{\frac{\pi}{2}} \times \frac{1}{m} \sum_{i=1}^m |z_{ij}^k - \bar{z}_j^k| & (12-2) \\
 (He)_j^k &= \sqrt{(S_k)^2 - ((En)_j^k)^2} & (12-3) \\
 (S_k)^2 &= \frac{1}{m-1} \sum_{i=1}^m (z_{ij}^k - \bar{z}_j^k)^2 & (12-4) \\
 \sum_{j=1}^n \omega_j^k &= 1 & (12-5) \\
 \omega_j^k &\geq 0 & (12-6) \\
 j &= 1, 2, \dots, n, k = 1, 2, \dots, s & (12-7)
 \end{aligned} \right. \quad s.t. \quad (12)
 \end{aligned}$$

where, ω_j^k ($j = 1, 2, \dots, n, k = 1, 2, \dots, s$) represents the weight of attribute c_j ($j = 1, 2, \dots, n$) for decision maker d_k ($k = 1, 2, \dots, s$). The evaluation information z_{ij}^k ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$) is the input variable.

The objective function of the model is to minimize the comprehensive uncertainty of each attribute, so as to determine a reasonable attribute weight. The constraints (12-1)–(12-4) are for calculating the parameters of the cloud model. The constraint condition (12-5) ensures the normalization of the weights. Constraints (12-6)–(12-7) constrain the value range of each parameter.

Theorem 5. The model M_1 has an optimal solution.

Proof. The objective function of model M_1 is defined as $f(\omega^k) = \sum_{j=1}^n \omega_j^k \cdot (He)_j^k$, where $(He)_j^k$ represents the hyper entropy of attribute c_j , which is calculated based on the evaluation information z_{ij}^k provided by each decision maker and is treated as a known constant. Therefore, the objective function is a linear function with respect to the attribute weight vector $\omega^k = (\omega_1^k, \dots, \omega_n^k)^T$.

For each variable ω_j^k , the first-order partial derivative is given by $\frac{\partial f}{\partial \omega_j^k} = (He)_j^k$. The second-order partial derivatives are all zero, that is, the Hessian matrix $H = 0_{n \times n}$, which is positive semi-definite, indicating that the objective function is convex. The constraints include the normalization condition $\sum_{j=1}^n \omega_j^k = 1$ and the non-negativity condition $\omega_j^k \geq 0$, which together form a standard probability simplex. This feasible region is both closed and convex.

In conclusion, according to convex optimization theory, model M_1 has an optimal solution. $\square$

4.3. A group consensus model considering cooperative behavior of decision makers based on fairness theory

In the process of group decision making, there are usually significant differences in the initial evaluation and preferences of different decision makers, especially in the context of MAGDM, each decision maker needs to receive feedback from the top decision makers based on group preferences, as well as making judgments based on their own evaluations. Such feedback often requires decision makers to adjust their initial evaluation in order to form a unified decision-making direction with the group. A traditional group consensus decision model is as follows:

$$M_2 : \min \sum_{k=1}^s \sum_{i=1}^m \sum_{j=1}^n |z_{ij}^k - \hat{z}_{ij}^k| \quad (13-1)$$

$$s.t. \begin{cases} \hat{z}_{ij}^0 = \sum_{k=1}^s \lambda_k^0 z_{ij}^k & (13-1) \\ \sum_{k=1}^s \lambda_k \left(1 - \frac{\sum_{i=1}^m \sum_{j=1}^n |z_{ij}^k - \hat{z}_{ij}^k|}{mn} \right) = CL^* & (13-2) \\ \hat{z}_{ij}^k \in [0, 1] & (13-3) \\ i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s & (13-4) \end{cases} \quad (13)$$

where, CL^* ($0 \leq CL^* \leq 1$) represents the maximum group consensus level, and $\hat{z}_{ij}^k$ ($i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, s$) represents the feedback suggestion value. The constraints (13-1)–(13-3) measure the group consensus level.

However, relying solely on feedback suggestions to adjust decisions can ignore the independent contribution and perception of fairness of individual decision makers. If a decision maker perceives the feedback as inconsistent with their expectations, or even perceives it as unfair, the decision maker may not fully accept the adjustment, even if it is within their confidence range. This perceived unfairness can affect the reaching of group consensus and reduce the efficiency of decision-making. Therefore, model (14) considers the confidence constraint on the basis of model (13), as follows:

$$M_3 : \min \left(\sum_{k=1}^s \sum_{i=1}^m \sum_{j=1}^n |z_{ij}^k - \hat{z}_{ij}^k| + \varepsilon \right) \quad (14)$$

$$s.t. \begin{cases} \varepsilon = \eta \sum_{i=1}^m \sum_{j=1}^n \max \left\{ 0, |z_{ij}^k - \hat{z}_{ij}^k| - b_j^k \right\} & (14-1) \\ \text{The constraints in model (13)} & (14-2) \end{cases}$$

where, the constraint (14-1) defines a penalty term ε that represents the additional adjustment cost incurred when the difference between the decision maker's initial evaluation and the feedback proposal exceeds the confidence range, so as to avoid over-reliance on some extreme individual evaluation when the group reaches a consensus, thereby maintaining the balance of the model. b_j^k ($j = 1, 2, \dots, n, k = 1, 2, \dots, s$) represents the degree of confidence that decision maker d_k ($k = 1, 2, \dots, s$) places on attribute c_j ($j = 1, 2, \dots, n$).

In addition, in order to deal with the problem of multi-party interest conflict in complex decision-making environment and the challenge of difficult to form consensus in group decision-making process, cooperative game theory provides an effective solution. Cooperative game theory emphasizes that each participant achieves the goal of maximizing the overall benefit through cooperation, and ensures that each participant's contribution can be measured and compensated fairly through reasonable distribution mechanism. As an important allocation method, Shapley value (Liu & Zhu, 2021; Zhao et al., 2023) can effectively measure the marginal contribution of each decision maker in collective

decision-making. Therefore, model (14) is extended to model (15).

$$M_4 : \min \left(\phi_k \sum_{k=1}^s \sum_{i=1}^m \sum_{j=1}^n |z_{ij}^k - \hat{z}_{ij}^k| + \varepsilon \right) \quad (15-1)$$

$$s.t. \begin{cases} \phi_k = \sum_{S \subseteq D \setminus \{d_k\}} \frac{|S|!(|D|-|S|-1)!}{|D|!} (v(S \cup \{d_k\}) - v(S)) & (15-1) \\ v(S) = \sum_{d_k \in S} \sum_{j=1}^n \omega_j^k z_{ij}^k f(z_{ij}^k) = \sum_{d_k \in S} \sum_{j=1}^n \frac{\omega_j^k z_{ij}^k}{1 + \gamma \cdot (z_{ij}^k)^\delta} & (15-2) \\ \text{The constraints in model (14)} & (15-3) \end{cases} \quad (15)$$

where, the constraint (15-1) is a formula for calculating the Shapley value ϕ_k ($k = 1, 2, \dots, s$), which measures the marginal contribution of each decision maker d_k ($k = 1, 2, \dots, s$) throughout the decision making process. Constraint (15-2) is a definition of the utility function used to calculate the total utility contribution of the decision maker in the group.

The group consensus model effectively incorporates the risk aversion characteristics of the decision maker into the decision-making process by embedding the risk aversion function in the utility function. Specifically, the utility function combines the risk avoidance feature and reflects the different sensitivity of decision makers to risk by adjusting the attribute utility value. The risk avoidance function dynamically weakens the utility contribution of high-risk attributes according to the perceived risk value of attributes and the decision-maker's risk sensitive parameters. The higher the perceived value of risk, the more significant the attenuation effect of risk avoidance function, indicating that decision makers show a stronger tendency to avoid in high-risk situations, while maintaining a higher utility contribution to low-risk attributes. This mechanism enables utility functions not only to reflect the importance of attributes, but also to capture decision-makers' subjective attitudes toward different levels of risk. On this basis, the results calculated by the utility function are further introduced into the group consensus model to achieve the balance between individual preference and group consistency by optimizing the objective function.

On this basis, this paper constructs an optimization model to obtain the feedback and suggestion matrix of each decision maker by minimizing the adjustment cost of the group and taking into account the contribution rate and fairness perception of individual decision makers. The model constructed in this paper measures each decision maker's contribution to group decision making based on Shapley value, and introduces fairness perception adjustment factor to ensure that decision makers' adjustment behavior can reflect their fairness perception of feedback. In addition, the risk avoidance level of decision makers is included in the utility function to reflect the influence of decision makers' risk preference on their judgment. The model is as follows:

$$M_5 : \min \sum_{k=1}^s \left(\phi_k \cdot \sum_{i=1}^m \sum_{j=1}^n C_k + \varepsilon \right) \quad (16-1)$$

$$s.t. \begin{cases} C_k = \left| \left(z_{ij}^k - \hat{z}_{ij}^k \right) - \frac{\alpha_k}{s-1} \sum_{k' \neq k} \max \left\{ z_{ij}^k - \hat{z}_{ij}^k, 0 \right\} \right| & (16-1) \\ \left| \frac{\chi_k}{s-1} \sum_{k' \neq k} \max \left\{ z_{ij}^k - \hat{z}_{ij}^k, 0 \right\} \right| & (16-2) \\ \text{The constraints in model (15)} & (16-2) \end{cases} \quad (16)$$

where ε represents the total cost of additional adjustments, η ($\eta \geq 0$) is the additional unit adjustment cost, α_k ($0 \leq \alpha_k \leq 1$) represents the jealousy coefficient, χ_k ($-1 \leq \chi_k \leq 1$) represents the sympathy coefficient.

The objective function of the model is to minimize the adjustment cost of decision makers in the process of group decision making, while taking into account the contribution rate of each decision maker and their perception of fairness. Constraint condition (16-1) is the definition of fairness perception factor, that is, in the process of group consensus, the fairness perceived by decision makers to the feedback provided by the top decision maker will affect their adjustment behavior.

The optimal solution structure of model M_5 is basically the same as that of model M_1 , and the proof process of the two is highly consistent. In order to avoid the repetition of theorems and proofs, the existence proof of the optimal solution of this model can refer to the derivation structure of model M_1 .

4.4. Quantum probability computation based on quantum-like Bayesian networks

In the traditional decision-making model, the interaction between decision makers is often simplified, and it is difficult to reflect the nonlinear and non-independent psychological interference effect. As a fusion form of quantum probability theory and Bayesian structure, quantum-like Bayesian networks can model such non-classical uncertainties by introducing interference terms while retaining probabilistic reasoning logic (Moreira & Wichert, 2016, 2018). Therefore, this paper constructs quantum-like Bayesian networks to deal with interference effects among decision makers. Among them, through the construction of conditional probability structure, the model can organically integrate the feedback information adjusted by group consensus with attribute weights, and convert it into quantized preference expression, providing theoretical basis and data support for the vector modeling of subsequent interference angles and the introduction of interference effects.

4.4.1. Construct quantum-like Bayesian networks

Firstly, the weights of decision makers are transformed into the first layer of quantum-like Bayesian networks.

$$P(d_k) = \lambda_k, k = 1, 2, \dots, s \quad (17)$$

Secondly, the conditional probability is calculated by weighting the feedback suggestions of each decision maker and the weight of the attribute.

$$P(a_i|d_k) = \frac{\sum_{j=1}^n \omega_j^k z_{ij}^k}{\sum_{i=1}^m \sum_{j=1}^n \omega_j^k z_{ij}^k}, i = 1, 2, \dots, m, k = 1, 2, \dots, s \quad (18)$$

4.4.2. Phase angle calculation of interference term based on similar heuristic algorithm

In this paper, a similar heuristic algorithm (Moreira et al., 2020) is used to calculate the interference angle between decision makers. Firstly, construct vectors $\vec{v}_k (k = 1, 2, \dots, s)$ and $\vec{\delta}_h (h = 1, 2, \dots, s)$.

$$\vec{v}_k = \begin{bmatrix} P(d_k)P(a_1|d_k) \\ P(d_k)P(a_2|d_k) \\ \vdots \\ P(d_k)P(a_m|d_k) \end{bmatrix}, \vec{\delta}_h = \begin{bmatrix} P(d_h)P(a_1|d_h) \\ P(d_h)P(a_2|d_h) \\ \vdots \\ P(d_h)P(a_m|d_h) \end{bmatrix} \quad (19)$$

$k = 1, 2, \dots, s, h = 1, 2, \dots, s, h \neq k$

On this basis, the deviation of vectors $\vec{v}_k (k = 1, 2, \dots, s)$ and $\vec{\delta}_h (h = 1, 2, \dots, s)$ is defined as:

$$\|\vec{\Delta}\| = \|\vec{v}_k - \vec{\delta}_h\| \quad (20)$$

Secondly, the angle between each vector is calculated, that is, the interference angle.

$$\begin{aligned} \theta_{\vec{v}_k} &= \cos^{-1} \left(\frac{\|\vec{\delta}_h\| + \|\vec{\Delta}\| - \|\vec{v}_k\|}{2 \cdot \|\vec{\delta}_h\| \cdot \|\vec{\Delta}\|} \right), k = 1, 2, \dots, s, \\ \theta_{\vec{\delta}_h} &= \cos^{-1} \left(\frac{\|\vec{v}_k\| + \|\vec{\Delta}\| - \|\vec{\delta}_h\|}{2 \cdot \|\vec{v}_k\| \cdot \|\vec{\Delta}\|} \right), h = 1, 2, \dots, s, \\ \theta_{\vec{\Delta}} &= \cos^{-1} \left(\frac{\|\vec{v}_k\| + \|\vec{\delta}_h\| - \|\vec{\Delta}\|}{2 \cdot \|\vec{v}_k\| \cdot \|\vec{\delta}_h\|} \right). \end{aligned} \quad (21)$$

The similarity of interference information is defined as:

$$\mu_{kh} = \frac{\theta_{\vec{\Delta}} - \theta_{\vec{\delta}_h}}{\theta_{\vec{v}_k}}, k = 1, 2, \dots, s, h = 1, 2, \dots, s, k \neq h \quad (22)$$

Finally, determine the phase angle degree of the interference term:

$$\theta_{kh} = \begin{cases} 0, & \mu \leq -2; \\ \frac{\pi}{2} (1 + 0.5 \cdot \mu), & -2 < \mu < 2; \\ \pi, & \mu \geq 2; \end{cases} \quad (23)$$

$k = 1, 2, \dots, s, h = 1, 2, \dots, s, k \neq h.$

On this basis, the quantum probability values of each alternative are calculated.

$$\begin{aligned} P(a_i) &= \left| \sum_{k=1}^s \varphi_{d_k} \varphi_{a_i|d_k} e^{i\theta_k} \right|^2 \\ &= \frac{1}{\rho} \left[\sum_{k=1}^s P(d_k) P(a_i|d_k) \right. \\ &\quad \left. + 2 \sum_{k=1}^{s-1} \sum_{h=k+1}^s \sqrt{P(d_k) P(a_i|d_k) P(d_h) P(a_i|d_h)} \cos(\theta_{kh}) \right] \quad (24) \\ &= \frac{1}{\rho} \left[\sum_{k=1}^s P(d_k) P(a_i|d_k) + interference_i \right] \\ &\quad i = 1, 2, \dots, m \end{aligned}$$

where, the normalization factor ρ is defined as:

$$\rho = \sum_{i=1}^m \left(\sum_{k=1}^s P(d_k) P(a_i|d_k) + interference_i \right) \quad (25)$$

4.5. Decision steps

Step 1: Collect the initial evaluation matrix of each decision maker, and carry out normalization processing according to Eq. (10) on this basis.

Step 2: Calculate the optimal attribute weights according to model (12).

Step 3: According to model (16), the feedback and suggestion matrix of each decision maker is obtained.

Step 4: Construct a quantum-like Bayesian network and calculate conditional probabilities according to (18).

Step 5: Calculate and rank the quantum probabilities of each alternative according to Eqs. (19)–(25).

5. Case analysis

In this paper, case is analyzed based on real policy background. While the design process drew heavily on actual policy practice and consulted with relevant government departments, the expert rating data did not come from formal government reviews. Instead, it is used to verify the applicability and theoretical reasonableness of the proposed model. Therefore, this case does not attempt to reproduce the official decision-making process, but rather provides a context-driven test bed to demonstrate the explanatory power and applicability of the model in real-world complex decision-making environments.

5.1. Background

On August 16, 2024, Nanjing City hosted the “High-Quality Development of Intelligent Connected Automobile Industry National ‘Vehicle Road Cloud Integration’ Application Pilot Policy Conference”, during which the “Nanjing Intelligent Connected Automobile Industry High-Quality Development Action Plan (2024–2026)” was officially unveiled. According to the plan, Nanjing will gradually open road tests and demonstration applications for intelligent connected vehicles over the next three years. This initiative requires careful planning to promote driverless technology, ensuring that the intelligent connected vehicle system operates smoothly in various traffic environments, enhances traffic safety, and optimizes travel efficiency. The city’s Transportation Group also announced plans to build a unified

Table 1
Initial evaluation information matrix $Z'^k (k = 1, 2, \dots, 5)$ for each decision maker.

	c_1	c_2	c_3	c_4	c_5	c_1	c_2	c_3	c_4	c_5	c_1	c_2	c_3	c_4	c_5
	d_1					d_2					d_3				
a_1	4	3	3	7	2	5	3	10	7	1	4	1	8	1	8
a_2	9	5	6	8	4	2	9	2	6	7	6	3	5	6	3
a_3	5	2	7	9	7	10	7	4	3	4	3	4	4	2	2
a_4	3	1	5	6	2	7	2	5	4	10	4	7	6	9	7
a_5	7	6	6	4	9	4	5	9	5	8	9	2	9	2	10
	d_4					d_5									
a_1	8	4	6	1	9	2	1	7	8	8					
a_2	7	6	8	2	8	5	3	8	6	3					
a_3	8	3	1	8	3	1	3	5	3	1					
a_4	4	1	5	10	7	6	10	6	4	9					
a_5	10	9	7	9	8	10	8	3	7	4					

“car-road-cloud” intelligent network vehicle control platform by year-end, which will coordinate “one cloud” management for road vehicles and support intelligent traffic management.

To steadily promote this strategic initiative, the Nanjing Municipal Government has outlined several alternatives, including: (1) Alternative a_1 : Priority deployment of unmanned systems on main roads. (2) Alternative a_2 : Pilot unmanned vehicles on certain secondary roads. (3) Alternative a_3 : Driverless testing in specific closed areas. (4) Alternative a_4 : Promote the use of driverless vehicles in public transport. (5) Alternative a_5 : Explore the coexistence of driverless and conventional vehicles.

In this context, Nanjing invited decision makers from traffic management, driverless technology, economics, sociology, and environmental protection to participate in the decision-making process. Experts evaluated each program based on their professional expertise in their respective fields. During the evaluation, decision makers assessed each option across five key attributes. Each attribute is rated on a scale from 0 to 10, where 0 indicates minimal compliance or satisfaction and 10 indicates full compliance or the highest satisfaction. Among them, the weight of each decision maker is evenly distributed. The detailed explanations and scoring criteria for the five attributes are as follows:

(1) Implementation cost c_1 : This attribute measures the investment required to fully implement intelligent connected vehicles in Nanjing, including infrastructure transformation, equipment installation, technical support, and maintenance costs.

(2) Technical feasibility c_2 : This attribute evaluates the maturity and implementation difficulty of the required technology. It considers the applicability of existing technology, the complexity of implementation, and technical resource availability.

(3) Security c_3 : This attribute reflects the solution's ability to protect unmanned vehicles and their surrounding environment in various scenarios, including reliability in collision avoidance, data security, and emergency response capabilities.

(4) Traffic flow improvement potential c_4 : This attribute evaluates the scheme's potential to improve overall traffic flow in Nanjing, especially in alleviating congestion during peak hours and improving traffic efficiency.

(5) Public acceptance c_5 : This attribute measures the level of support and approval from Nanjing citizens for the program. It reflects the social adaptability of the scheme and public response, including citizens' trust in driverless technology and recognition of the scheme's impact.

5.2. Decision steps

Step 1: Collect the initial evaluation information matrix $Z'^k (k = 1, 2, \dots, 5)$ for decision makers, as shown in Table 1.

Step 2: Normalize the initial evaluation information matrix of each decision maker according to Eq. (10), as shown in Table 2.

On this basis, the optimal attributes' weights of each decision maker are solved according to model (12), as shown in Table 3.

Step 3: The jealousy coefficient of each decision maker is considered as:

$$\alpha_1 = 0.3, \alpha_2 = 0.3, \alpha_3 = 0.2, \alpha_4 = 0.5, \alpha_5 = 0.1.$$

The sympathy coefficient of each decision maker was set as:

$$\chi_1 = 0.4, \chi_2 = 0.6, \chi_3 = 0.5, \chi_4 = 0.2, \chi_5 = 0.1.$$

The confidence $b_j^k (j = 1, 2, \dots, 5, k = 1, 2, \dots, 5)$ of each decision maker for each attribute is 0.1, the risk sensitivity coefficient γ is 0.5, and the nonlinearity degree parameter δ is 1.5. On the basis, solve the model (16) to obtain the feedback and suggestion matrix of each decision maker, as shown in Table 4.

Step 4: Construct a quantum-like Bayesian network, assume that the weight of decision makers is evenly distributed, and calculate the conditional probability according to Eq. (18) as follows:

$$\begin{aligned} P(a_1|d_1) &= 0.0586, & P(a_2|d_1) &= 0.2714, & P(a_3|d_1) &= 0.2444 \\ P(a_4|d_1) &= 0.1636, \\ P(a_5|d_1) &= 0.2620, & P(a_1|d_2) &= 0.2240, & P(a_2|d_2) &= 0.1405 \\ P(a_3|d_2) &= 0.3317, \\ P(a_4|d_2) &= 0.1282, & P(a_5|d_2) &= 0.1756, & P(a_1|d_3) &= 0.1333, \\ P(a_2|d_3) &= 0.1977, \\ P(a_3|d_3) &= 0.2647, & P(a_4|d_3) &= 0.2759, & P(a_5|d_3) &= 0.1284, \\ P(a_1|d_4) &= 0.2571, \\ P(a_2|d_4) &= 0.2449, & P(a_3|d_4) &= 0.1150, & P(a_4|d_4) &= 0.0654, \\ P(a_5|d_4) &= 0.3176, \\ P(a_1|d_5) &= 0.2365, & P(a_2|d_5) &= 0.2210, & P(a_3|d_5) &= 0.0561, \\ P(a_4|d_5) &= 0.2699, \\ P(a_5|d_5) &= 0.2164. \end{aligned}$$

Step 5: Calculate the quantum probability of each alternative according to the Eqs. (19)–(25):

$$\begin{aligned} P(a_1) &= 0.1818, & P(a_2) &= 0.2313, \\ P(a_3) &= 0.1802, & P(a_4) &= 0.1749, \\ P(a_5) &= 0.2318. \end{aligned}$$

The ranking results are as follows: $a_5 > a_2 > a_1 > a_3 > a_4$.

5.3. Sensitivity analysis

This section analyzes the sensitivity of several core parameters involved in this paper, and observes the influence of each parameter on the group consensus decision-making process. For the convenience of analysis, only one round of adjustment is considered, that is, the decision maker accepts the feedback and suggestion matrix given by the top decision maker. On this basis, sensitivity analysis is carried out in the following three aspects: (1) The change of risk sensitivity of each decision maker. (2) Changes in the confidence degree of each decision maker. (3) Changes in the perceived level of fairness among decision-makers.

(1) The impact of decision makers' risk sensitivity γ on group consensus level

In multi-attribute group decision making, the risk sensitivity of decision makers significantly affects the process of reaching group consensus. In order to study this effect, we assume that all decision makers have the same risk sensitivity, and analyze the impact of sensitivity change on group consensus and group adjustment cost. Fig. 4 shows the effect of different sensitivities on the group consensus process.

As can be observed from Fig. 4, the group consensus level shows significant fluctuations as risk sensitivity increases. When the risk sensitivity is low (0.1 to 0.3), the group consensus level is relatively low. The group consensus reached a high level when the risk sensitivity was moderate (around 0.7). However, in areas with risk sensitivity

Table 2
Normalized decision matrix $Z^k(k = 1, 2, \dots, 5)$ for each decision maker.

	c_1	c_2	c_3	c_4	c_5		c_1	c_2	c_3	c_4	c_5
	d_1						d_2				
a_1	0.1667	0.4000	0.0000	0.4000	1.0000		0.3750	0.1429	1.0000	0.0000	1.0000
a_2	1.0000	0.8000	0.7500	0.2000	0.7143		0.0000	1.0000	0.0000	0.2500	0.3333
a_3	0.3333	0.2000	1.0000	0.0000	0.2857		1.0000	0.7143	0.2500	1.0000	0.6667
a_4	0.0000	0.0000	0.5000	0.6000	1.0000		0.6250	0.0000	0.3750	0.7500	0.0000
a_5	0.6667	1.0000	0.7500	1.0000	0.0000		0.2500	0.4286	0.8750	0.5000	0.2222
	d_3						d_4				
a_1	0.1667	0.0000	0.8000	1.0000	0.2500		0.6667	0.3750	0.7143	1.0000	0.0000
a_2	0.5000	0.3333	0.2000	0.3750	0.8750		0.5000	0.6250	1.0000	0.8889	0.1667
a_3	0.0000	0.5000	0.0000	0.8750	1.0000		0.6667	0.2500	0.0000	0.2222	1.0000
a_4	0.1667	1.0000	0.4000	0.0000	0.3750		0.0000	0.0000	0.5714	0.0000	0.3333
a_5	1.0000	0.1667	1.0000	0.8750	0.0000		1.0000	1.0000	0.8571	0.1111	0.1667
	d_5										
a_1	0.1111	0.0000	0.8000	0.0000	0.1250						
a_2	0.4444	0.2222	1.0000	0.4000	0.7500						
a_3	0.0000	0.2222	0.4000	1.0000	1.0000						
a_4	0.5556	1.0000	0.6000	0.8000	0.0000						
a_5	1.0000	0.7778	0.0000	0.2000	0.6250						

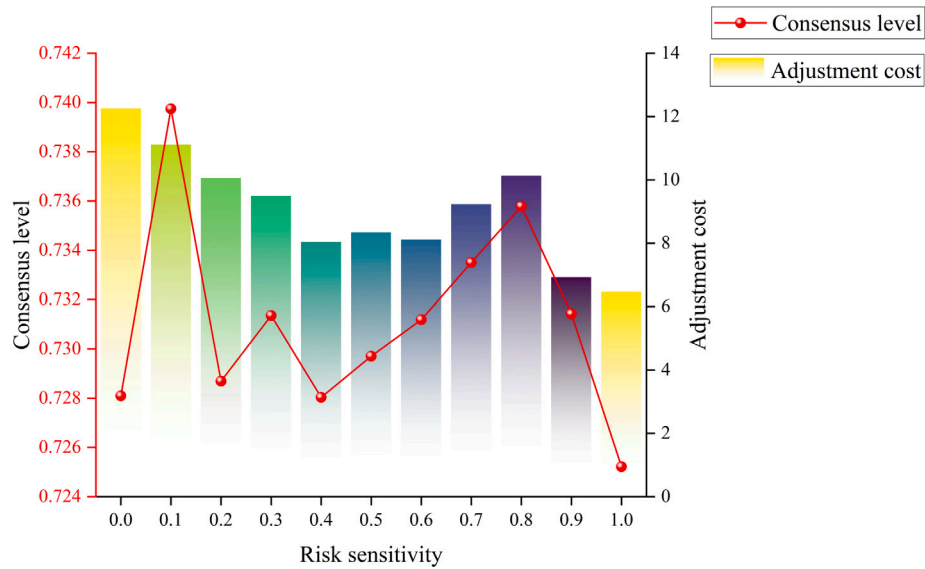


Fig. 4. Influence of risk sensitivity of decision makers on group decision making.

Table 3
The weights of the optimal attributes for each decision maker.

	c_1	c_2	c_3	c_4	c_5
d_1	0.2338	0.0383	0.4669	0.1732	0.0878
d_2	0.1099	0.2443	0.0167	0.2897	0.3394
d_3	0.1131	0.5049	0.0540	0.1964	0.1316
d_4	0.2944	0.3918	0.1084	0.1696	0.0358
d_5	0.2122	0.1238	0.0810	0.2293	0.3537

approaching 1, the consensus level decreases significantly. In addition, the adjustment cost also shows a significant trend with the change of risk sensitivity. In the region of high risk sensitivity (0.7 to 1.0), the adjustment cost showed a gradual decline, while in the region of low and medium risk sensitivity, the adjustment cost was relatively high. This shows that different risk attitudes will affect the coordination difficulty and overall cost of decision making.

Low risk sensitivity often implies a higher acceptance of risk by individuals, but it can also make it difficult for group members to reach consensus on some of the more conservative options, resulting in a lower level of consensus. At the same time, this lower risk sensitivity may increase the divergence of parties in decision-making, thus

increasing the cost of adjustment. When risk sensitivity is moderate, group members are more likely to find the balance point and reach a higher consensus, which reflects the importance of moderate risk tolerance for group consistency. However, when risk sensitivity is too high, adjustment costs are also significantly reduced despite reduced consensus levels, possibly because highly risk-sensitive individuals tend to compromise and seek safety strategies that avoid high-risk scenarios.

In group decision making, maintaining a moderate risk sensitivity helps to balance the risk attitude among individuals, promote consensus and reduce the cost of adjustment. Especially in the multi-party decision-making environment with significant conflicts of interest and high complexity, adjusting risk sensitivity can improve the effectiveness and consistency of group decision-making.

(2) The influence of decision maker's confidence level b_j^k ($j = 1, 2, \dots, 5, k = 1, 2, \dots, 5$) on group decision making

In the complex case of multi-attribute group decision making, confidence, as the certainty of individual decision making, has a crucial impact on group consensus and decision adjustment cost. Reasonable trust regulation can effectively balance individual confidence in decision-making and openness to other members' opinions. This paper assumes that the confidence of all decision makers is the same, and

Table 4
Feedback suggestion matrix $\hat{Z}^k (k = 1, 2, \dots, 5)$ of each decision maker.

	c_1	c_2	c_3	c_4	c_5		c_1	c_2	c_3	c_4	c_5
	d_1						d_2				
a_1	0.0415	0.2068	0.0307	0.2099	0.9760		0.4650	0.3482	0.9953	0.0966	0.9728
a_2	0.9366	0.3530	0.7789	0.2445	0.8544		0.2920	0.2113	0.1922	0.5274	0.2381
a_3	0.6430	0.4890	0.9444	0.0514	0.2695		0.9813	0.8874	0.2095	0.6937	0.6707
a_4	0.1477	0.0358	0.3752	0.7657	0.9882		0.6584	0.1305	0.2400	0.5732	0.0534
a_5	0.6770	0.7960	0.7049	0.9156	0.1408		0.2901	0.4859	0.7655	0.6913	0.1092
	d_3						d_4				
a_1	0.0633	0.0337	0.8860	0.9573	0.1512		0.6889	0.3750	0.8145	0.8436	0.5987
a_2	0.0955	0.4066	0.2222	0.3554	0.8917		0.2609	0.6304	0.6281	0.9651	0.5085
a_3	0.2354	0.5081	0.1438	0.7805	0.8506		0.5565	0.1605	0.1087	0.0487	0.6351
a_4	0.6064	0.8032	0.4279	0.0678	0.5255		0.0185	0.1414	0.6648	0.0331	0.4094
a_5	0.4794	0.0353	0.7964	0.7487	0.0580		0.9072	0.9312	0.7207	0.1413	0.2861
	d_5										
a_1	0.1035	0.0343	0.9284	0.9213	0.9257						
a_2	0.4872	0.3412	0.8084	0.8047	0.5726						
a_3	0.0371	0.1565	0.2403	0.2199	0.1549						
a_4	0.4010	0.9690	0.6419	0.5548	0.9791						
a_5	0.9825	0.8614	0.6331	0.9002	0.0363						

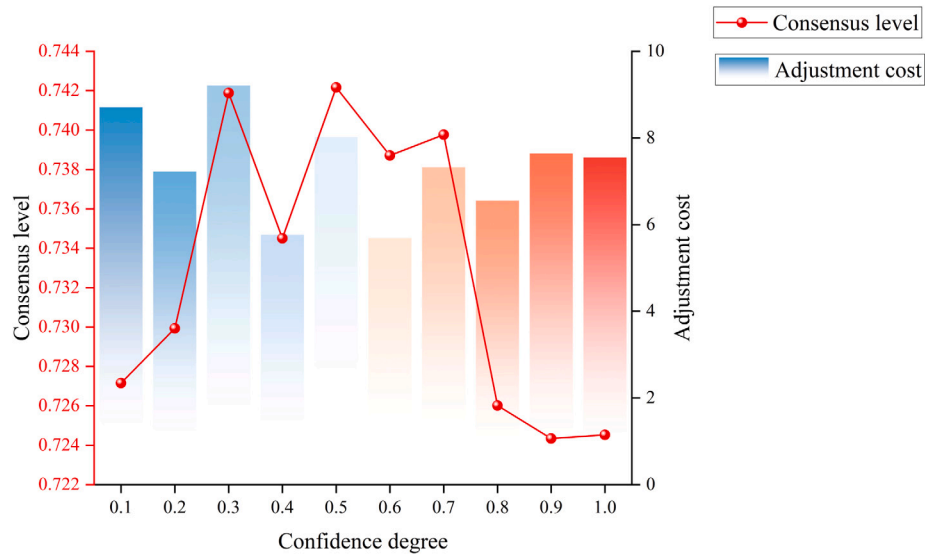


Fig. 5. Sensitivity analysis of each decision maker's confidence level to group decision making.

observes the dynamic impact of its change on the group consensus level and adjustment cost, as shown in Fig. 5.

As can be seen from Fig. 5, with the increase of confidence, the group consensus level presents significant volatility. For example, group consensus is higher at medium levels of confidence (around 0.3 and 0.7), but lower at very high or very low levels of confidence. In addition, the adjustment cost also shows different trends with the change of confidence degree. With the increase of confidence degree, the adjustment cost generally presents a trend of fluctuation and increase, especially when the confidence degree is high (0.8 to 1.0), the adjustment cost increases significantly. In contrast, medium values of confidence (such as 0.3 to 0.7) promote high consensus among group members, while too high or too low confidence lowers consensus levels.

This is because the low confidence level leads to the lack of mutual trust between individuals, making it difficult to form a consensus. However, a high degree of confidence may lead to overconfidence in one's own judgment and reduce the acceptance of other opinions, thus affecting the group consistency. At the same time, when the confidence is low, the adjustment cost is small, and when the confidence is gradually increased, the adjustment cost is also significantly increased. This indicates that in the process of group decision making, although higher confidence may increase individuals' confidence in their own decisions,

it will also make compromise and adjustment more difficult, thus increasing the overall adjustment cost. Therefore, in group decision-making, a moderate level of confidence is essential to reach an effective group consensus. Keeping the confidence level within the moderate range not only helps to improve the consensus level of the group, but also helps to control the adjustment cost, so as to achieve more efficient and robust group decision-making.

(3) The influence of decision maker's perception of fairness on group decision making

For complex and uncertain multi-attribute group decision making, fairness effect theory points out that individual emotional factors (such as jealousy and sympathy) in the decision-making process can significantly affect the group's overall utility and the formation of consensus. Therefore, analyzing the sensitivity of jealousy coefficient α_k ($0 \leq \alpha_k \leq 1$) and sympathy coefficient χ_k ($-1 \leq \chi_k \leq 1$) can more deeply reveal the mechanism of action of these emotional responses in group decision-making, as shown in Figs. 6 and 7.

Fig. 6 shows the effect of coefficient of jealousy α_k ($0 \leq \alpha_k \leq 1$) and coefficient of sympathy χ_k ($-1 \leq \chi_k \leq 1$) on the group consensus level. When the sympathy coefficient is high ($\chi_k > 0.5$) and the jealousy coefficient is low, the group consensus level is high. This means that increasing empathy in situations where reducing negative emotions

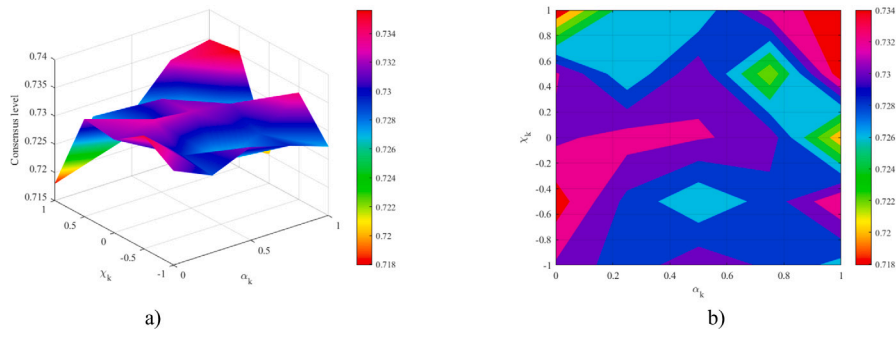


Fig. 6. Sensitivity analysis of fairness utility to group consensus level.

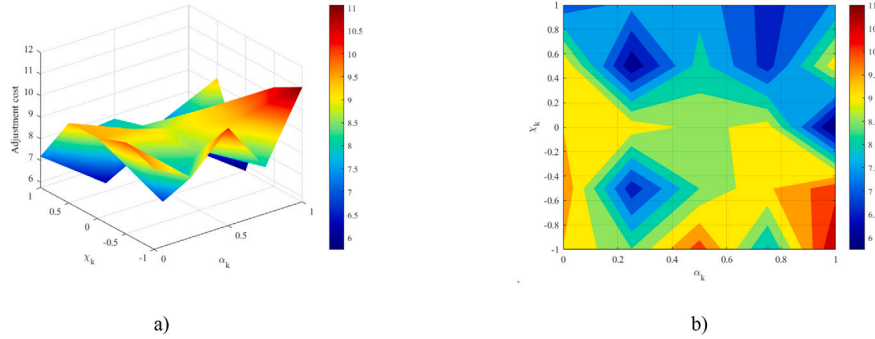


Fig. 7. Sensitivity analysis of equity utility to adjustment cost in group consensus process.

toward others with higher benefits helps to increase the level of group consensus. However, when the jealousy coefficient is high ($\alpha_k > 0.5$), the consensus level shows a decreasing trend, especially when the sympathy coefficient is low.

Fig. 7 shows the effect of coefficient of jealousy α_k ($0 \leq \alpha_k \leq 1$) and coefficient of sympathy χ_k ($-1 \leq \chi_k \leq 1$) on the adjusted cost. The adjustment cost increases with the increase of jealousy coefficient, but decreases significantly when the sympathy coefficient is high ($\chi_k > 0$). Especially in the region where χ_k is high and α_k is low, the adjustment cost is at a relatively low level. This indicates that in the group decision-making process, increasing the sympathy coefficient can effectively reduce the adjustment cost, while higher jealousy coefficient will significantly increase the difficulty of coordination and adjustment.

In summary, a higher coefficient of sympathy with a lower coefficient of jealousy helps promote consensus among group members, thereby improving group consistency and reducing disagreement in decision making. At the same time, such a combination can also significantly reduce the adjustment cost, thus reducing the cost of the group in the process of achieving consensus. These results suggest that compassion is an important factor in promoting cooperation and reducing conflict in group decision making, while jealousy may hinder consensus formation and increase adjustment costs. Therefore, in the multi-interest conflict and complex decision-making environment, controlling the jealousy coefficient and improving the level of compassion can effectively improve the efficiency and consistency of group decision-making. These results highlight the importance of managing individual emotional responses in the decision-making process, especially when multiple parties have conflicting interests, and suggest optimizing the decision-making process by cultivating empathy among decision makers and reducing competitive negative emotions to achieve more efficient and robust group decision-making.

5.4. Simulation analysis I

The purpose of this simulation analysis is to construct five random decision matrices with different dimensions and input them into the

proposed model to study the evolution of the group consensus process under varying levels of complexity and to observe the changes in adjustment cost. To avoid the influence of specific data patterns on the results, each decision matrix is generated using MATLAB's "rand" function, thereby simulating numerical inputs in different decision-making scenarios. Although these data do not directly represent fuzzy information in practical applications, they mimic the transformed crisp values typically derived from fuzzy data such as linguistic terms or triangular fuzzy numbers. Since the conversion from fuzzy to crisp numbers is generally performed as a preprocessing step rather than within the model itself, this simulation setup effectively demonstrates the model's adaptability and robustness to different data types without affecting the structural integrity of the model. The corresponding simulation results are shown in Fig. 8.

According to the simulation results in Fig. 8, it can be seen that in the five rounds of adjustment rounds, the group consensus level shows a significant upward trend in the matrices of different sizes, and can all reach a relatively ideal level ($CL \geq 0.8$), this verifies that the model has a good adaptability and convergence ability to input disturbance, and the model can maintain an effective consensus evolution trend, and ensure the stability of the decision-making process, reflecting a strong robustness. However, with the increase of the dimensions of the decision matrix, the adjustment cost also increases significantly, indicating that in more complex group decision-making situations, organizations need to pay more adjustment costs in order to achieve a higher level of consensus. In addition, through the marginal contribution in the cooperative game, we can not only quantify the relative importance of each member in the overall decision making, but also assess the influence level of decision makers on the group consensus, that is, with the advancement of the decision-making process, the influence of individual opinions gradually emerges, thus promoting the improvement of the group consensus level.

The simulation results show that with the increase of decision rounds, the group consensus level gradually increases, indicating that the differences of opinions gradually narrow during the continuous

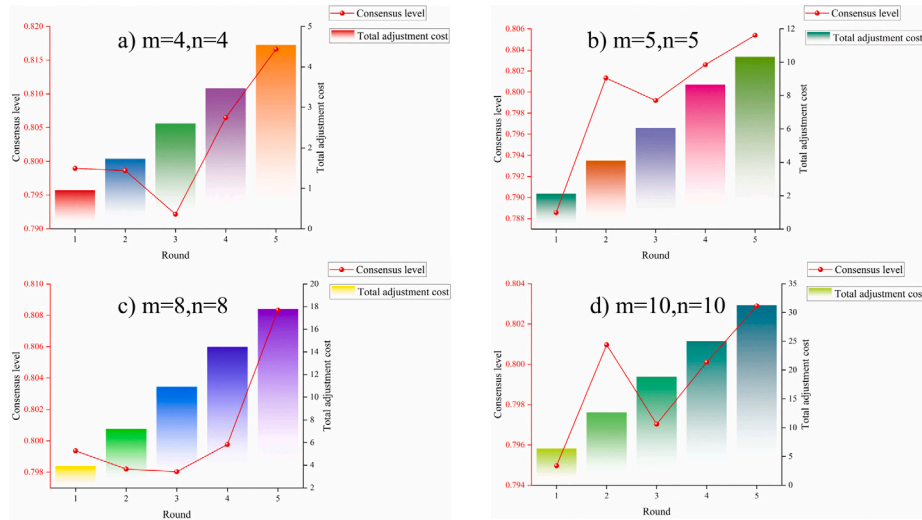


Fig. 8. Group consensus simulation evolution of decision matrices with different dimensions.

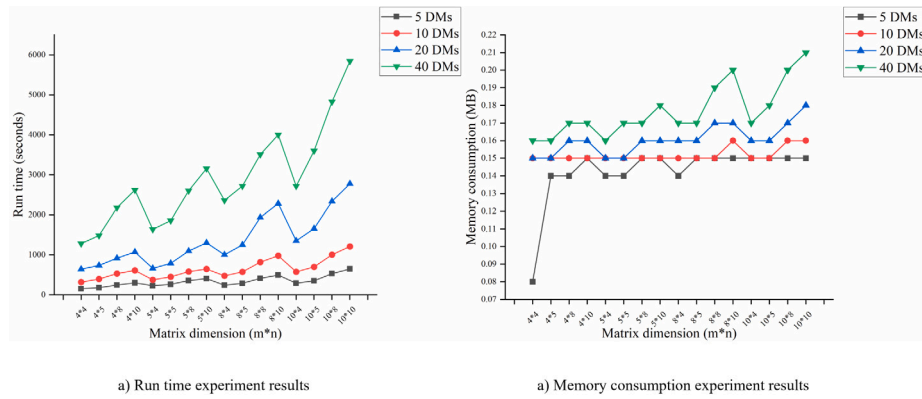


Fig. 9. Simulation experiment of computation cost of matrix dimension change.

interaction and adjustment process. At the same time, in order to meet the fairness demands of various decision-makers, especially respecting the will of each member, the top decision-maker may need to sacrifice some consensus level in practice. Finally, the level of risk aversion of decision makers also has an important impact on consensus. Because of the difference in the level of risk avoidance, it is difficult for groups to reach a consensus quickly in the decision-making process, which has a certain obstacle to the improvement of the consensus level.

5.5. Simulation analysis II

(1) Group consensus perspective

In order to evaluate the impact of the number of decision-makers, the number of attributes, and the number of alternatives on computational overhead, this study conducted an experimental analysis and systematically compared time consumption, memory usage, and runtime under different configurations. In this experiment, the initial decision matrix was randomly generated using MATLAB's rand function. Specifically, the weights of the decision-makers and the attributes followed a uniform distribution, and other parameters were assigned based on the examples in this paper. Processor configuration: 12th Gen Intel® Core™ i7-12700H 2.30 GHz, 64-bit operating system, 16 GB RAM. The experimental results are shown in Fig. 9.

As shown in Fig. 9, with the increase in the number of decision-makers, attributes, and alternatives, memory consumption and runtime show a significant upward trend. The increase in memory and time consumption is especially noticeable with the increase in the number of

decision-makers. This is due to the mutual influence between decision-makers, which leads to more computation steps and storage demands. Additionally, as the number of attributes and alternatives increases, each added attribute or alternative increases the computational and storage requirements, further driving up memory and time overhead. For example, when the matrix dimension increases to 10×10 , the increase in runtime and memory consumption becomes more significant, reflecting the nonlinear growth of computation with matrix size.

Furthermore, as the number of decision-makers, the number of attributes, and the number of alternatives increase, the time complexity of computational overhead shows linear growth. Specifically, the time complexity can be expressed as $O(sm n)$, which is proportional to the product of the number of decision-makers, alternatives, and attributes. With the increase in the number of decision-makers, the number and size of decision matrices the model needs to process also increase, leading to a higher computational load. Similarly, an increase in the number of attributes and alternatives means that more decision data need to be processed, thus increasing the demand for computational resources. The experimental results are consistent with the theoretical analysis, indicating that the computational overhead significantly rises as the number of decision-makers, alternatives, and attributes increases.

In conclusion, with the increase in the number of decision-makers, attributes, and alternatives, computational overhead significantly increases, with the number of decision-makers having the most significant impact on the computational overhead.

(2) Quantum interference perspective

Table 5
Qualitative comparison with existing methods.

Literature	Attributes' weights	Decision-makers' weights	Group consensus process	Fairness theory	Risk preference	Marginal contribution	Quantum interference effect
Tang and Liao (2024)	Known	×	✓	✓	×	×	×
Liu and Zhu (2022)	Unknown	Unknown	✓	✓	×	×	✓
Li et al. (2024)	×	Known	✓	×	×	×	×
Zhang and Li (2022)	×	Known	✓	×	×	×	×
Jing and Chao (2021)	×	Known	✓	✓	×	×	×
Du et al. (2022)	×	Known	✓	✓	×	×	×
Guo, Zhang et al. (2024)	×	Known	✓	✓	×	×	×
Liu et al. (2025)	×	Known	✓	✓	×	×	×
Gong et al. (2023)	Known	×	✓	✓	×	×	×
Liao et al. (2024)	Known	Unknown	✓	×	×	✓	×
Zha et al. (2022)	Known	Known	✓	×	×	×	×
Wu et al. (2021)	Known	Known	×	×	×	×	✓
Jiang and Liu (2022)	Unknown	Unknown	✓	×	×	×	✓
Wang, Liu et al. (2024)	Unknown	Unknown	×	×	×	×	✓
The proposed method	Unknown	Unknown	✓	✓	✓	✓	✓

Interference effects in MAGDM mainly arise from the interactions between decision makers. Since the interference effect is caused by the interaction between each pair of decision makers, the number of interference effects increases quadratically as the number of decision makers, s , increases. Specifically, for s decision makers, there are $s(s+1)/2$ interference effects, reflecting the interaction between each pair of decision makers. As the number of decision makers increases, the increase in interference effects becomes significant, directly affecting the stability of group decision-making. While increasing the number of attributes and alternatives expands the size of the decision matrix and increases computational complexity, these changes mainly affect the dimensions of the decision matrix and the distribution of weights, with little direct impact on the increase of interference effects between decision makers. In contrast, increasing the number of decision makers directly leads to more interference effects because each additional decision maker increases the mutual influence within the group decision.

In summary, the impact of the number of decision makers on interference effects is much greater than that of changes in the number of attributes and alternatives. Therefore, in modeling and optimization processes, particular attention must be given to the challenges posed by the increase in interference effects with the growing number of decision makers.

5.6. Comparative analysis

5.6.1. Qualitative contrastive analysis

In order to demonstrate the theoretical advantages of the method proposed in this paper more clearly, this paper makes a qualitative comparison with some existing group consensus methods. By constructing a multi-dimensional evaluation index system, the differences in modeling ideas and theoretical expansion among different methods are identified, further highlighting the superiority and applicability of the model proposed in this paper in dealing with uncertain and complex decision-making backgrounds, as shown in Table 5.

It can be seen from Table 5 that the existing group consensus models have the following problems:

The existing group consensus models have certain limitations in the calculation of attribute weights. Most models simplify the objective calculation of attribute weights or adopt the method of subjective assignment, making it difficult to reflect the actual role of attributes in complex and uncertain decision-making processes.

(2) Although existing studies have introduced fairness perception in group consensus modeling, such as through feedback adjustment mechanisms or setting acceptance intervals to enhance the fairness of the process and increase individuals' willingness to participate,

the related models generally have the following two main deficiencies: First, the marginal contribution of individuals in the outcome of group decision-making has not been systematically depicted, and their moderating responsibility and actual influence in the formation of consensus have been ignored. Second, the modeling and measurement of decision-makers' psychological perception are still relatively weak.

(3) The current group consensus model usually assumes that decision-makers are independent of each other and does not fully consider the possible interference effects among individuals, which may cause deviations and distortions in the model effect in a complex decision-making environment. On the other hand, although quantum decision-making theory shows great potential in consensus modeling, related studies have not fully considered the influence of fairness perception and marginal contribution on the group decision-making process in practical applications.

However, the model proposed in this paper not only considers the marginal contribution of decision makers, but also effectively integrates the fairness perception and interference effect, providing a more comprehensive and accurate decision-making framework for group consensus. In addition, by constructing quantum-like Bayesian networks, this paper further considers the interference effect among decision makers, so as to output more accurate and reliable alternative ranking results.

5.6.2. Quantitative contrastive analysis

(1) Comparison of alternative ranking results

To validate the effectiveness and applicability of the proposed model, several representative classical models are selected for comparison. All models are applied to the same dataset under identical input conditions to compute and analyze their respective ranking results. By examining the differences in rankings generated by each method when processing the same evaluation information, the comparative analysis highlights the advantages of the proposed model in addressing complex interference effects and psychological perception factors. The detailed results are shown in Table 6.

As can be seen from Table 6, compared with the traditional decision model, the alternative scheme ranking obtained by the proposed method is significantly different. For example, the optimal alternative obtained by TOPSIS (Kacprzak, 2020) method is a_2 , while the optimal alternative obtained in this paper is a_5 . Although the optimal alternative obtained by TODIM (Wang, Nhieu et al., 2024) method and GDM (Zha et al., 2022) is a_5 , the order of other schemes is not consistent with that obtained in this paper. The reason for this difference is that the interference effect among decision makers is not considered in the traditional model, which leads to the subjective influence of decision makers on alternative ranking.

However, compared with the quantum model, although Wu et al. (2022) concluded that the optimal alternative was a_5 , the ranking of the

Table 6
Comparison of numerical experimental ranking results of different models.

Method	Ranking index	Ranking
TOPSIS (Kacprzak, 2020)	$C_1^* = 0.6189, C_2^* = 0.6452,$ $C_3^* = 0.5257, C_4^* = 0.5842,$ $C_5^* = 0.4569.$	$a_2 > a_1 > a_4 > a_3 > a_5$
TODIM (Wang, Nhieu et al., 2024)	$\phi(a_1) = -0.0385, \phi(a_2) = -1.118,$ $\phi(a_3) = -4.1838, \phi(a_4) = -2.8675,$ $\phi(a_5) = 1.5646.$	$a_5 > a_1 > a_2 > a_4 > a_3$
GDM (Zha et al., 2022)	$Q(a_1) = 0.5165, Q(a_2) = 0.5149,$ $Q(a_3) = 0.3910, Q(a_4) = 0.4126,$ $Q(a_5) = 0.6410.$	$a_5 > a_1 > a_2 > a_4 > a_3$
Quantum Model 1 (She et al., 2021)	$P(a_1) = 0.2894, P(a_2) = 0.2637,$ $P(a_3) = 0.0293, P(a_4) = 0.1630,$ $P(a_5) = 0.2546.$	$a_1 > a_2 > a_5 > a_4 > a_3$
Quantum Model 2 (Wu et al., 2022)	$P(a_1) = 0.1157, P(a_2) = 0.2253,$ $P(a_3) = 0.2378, P(a_4) = 0.1450,$ $P(a_5) = 0.2762.$	$a_5 > a_3 > a_2 > a_4 > a_1$
The proposed model	$P(a_1) = 0.1818, P(a_2) = 0.2313,$ $P(a_3) = 0.1802, P(a_4) = 0.1749,$ $P(a_5) = 0.2318.$	$a_5 > a_2 > a_1 > a_3 > a_4$

Table 7
Comparison of group consensus efficiency with existing models.

Method	Group consensus adjusts the number of rounds		
	Round 1	Round 2	Round 3
Fairness preference group consensus model (Liu et al., 2025)	0.7196	0.8023	0.8564
Group consensus model (Zhang & Li, 2023)	0.7198	0.8599	0.9300
The proposed method	0.7666	0.8580	0.8618

other schemes was quite different from that in this paper, because this method failed to accurately calculate the interference value between decision makers on the one hand, and did not take into account the consensus preference reaching in the decision-making process on the other hand. The optimal ranking alternative obtained by the method of She et al. (2021) is a_1 . Although the method measures specific disturbance values, it does not consider the risk perception and fairness perception in the decision-making process, nor the volatility of attribute weights in uncertain decision-making environments. In this paper, a quantum group consensus model is proposed considering the psychological state of decision makers, which takes into account the risk avoidance level of decision makers in the uncertain decision-making process and the fairness perception of decision makers in the feedback suggestions in the group consensus process. Meanwhile, based on the normal cloud model, an optimization model with minimum fluctuation is constructed to solve the attribute weights. Therefore, the model in this paper is more suitable for the actual decision-making scenario, and has better applicability and robustness.

(2) Group consensus efficiency comparison

In order to verify the advantages of the multi-attribute quantum group consensus model proposed in this paper, it is compared with two existing group consensus models. The purpose of this section is to demonstrate the effectiveness and applicability of the proposed model in dealing with complex and uncertain decision environments, as shown in Table 7.

According to Table 7, Fairness preference group consensus model (Liu et al., 2025) considers the perception of fairness in the process of group consensus, and proposes the fairness utility function to optimize the adjustment behavior of decision makers. Although it helps to improve the sense of participation of decision makers and the fairness of decision-making process, However, this model does not fully consider the marginal contribution of decision makers, which may lead to the enthusiasm of high-contribution members frustrated by the mismatch between returns and inputs, and excessive averaging may reduce the efficiency of group decision-making. Therefore, the convergence rate of the model is slow, and it is difficult to reach a high consensus

level quickly. Although the consideration of fairness perception helps to enhance the participation of decision makers and the fairness of the decision-making process, the lack of comprehensive modeling of the interaction and influence of decision makers in complex decision-making environments leads to the limitations of this model in terms of efficient consensus reaching. The Group consensus model (Zhang & Li, 2023) reached a consensus level of 0.9300 after three rounds of adjustment, showing its advantage in efficiently realizing group consensus. However, this model does not consider the psychological perception, marginal contribution and mutual interference effect of decision makers, and only optimizes the weight and uses the feedback mechanism to improve the efficiency of decision making. Therefore, the model may lack comprehensiveness and adaptability when dealing with complex and uncertain decision-making backgrounds, especially when multi-source interference and psychological preferences are greatly affected.

Compared with Fairness preference group consensus model (Liu et al., 2025), the model proposed in this paper has obvious advantages in the convergence speed of Group consensus, although compared with the Group consensus model (Zhang & Li, 2023), the consensus level is slightly lower. But it can still achieve the ideal state. In addition, the model not only considers the marginal contribution of decision makers, but also effectively deals with the interference effect among decision makers after the group consensus is reached, and further improves the accuracy of the ranking results through optimization and adjustment. Therefore, the model presented in this paper shows high adaptability and practicability in the complex and changeable decision environment.

6. Conclusion

Considering the background of complex uncertain decision making, this paper proposes a quantum multi-attribute group consensus model considering the psychological perception of decision makers. Firstly, an optimization model based on the normal cloud model is constructed to solve the optimal attributes' weights by minimizing the volatility of

attributes' weights. Secondly, a group consensus model with minimum adjustment cost is proposed, which not only considers the marginal contribution of decision makers to guide cooperation, but also takes into account the influence of fairness perception and risk avoidance level on the decision-making process. Finally, we construct a quantum-like Bayesian network to measure the interference effect between decision makers by similar heuristic algorithm, and then calculate the quantum probability of each alternative.

The conclusions of this paper are as follows: Firstly, the introduction of fairness effect improves the fairness of decision-making process, but it may limit the rapid improvement of group consensus level in some cases, because it is necessary to find a balance between fairness and consistency. Secondly, the diversity of decision makers' risk aversion levels has a significant impact on the consensus level. Organizations need to find a reasonable compromise among different risk aversion levels to ensure that the needs of each decision maker are respected, which also increases the difficulty of reaching consensus. Finally, as the size and complexity of decisions increase, adjustment costs rise significantly, indicating that organizations need to invest more resources to maintain fairness and respect individual preferences in complex decision situations.

Although the quantum multi-attribute group consensus model proposed in this paper is innovative and effective in theory, however, there are some shortcomings. First of all, the complexity of the model is high, although it improves the decision-making accuracy, it also increases the computational cost and implementation difficulty. Especially when dealing with quantum interference effects in large-scale group decision-making, how to simplify the model or improve the computational efficiency still needs to be further studied. Secondly, although the quantum interference angle measurement method based on cosine similarity can effectively describe the mutual interference among decision makers, it may have shortcomings in computational accuracy and robustness in actual complex scenes. Further optimization of the measurement algorithm is needed in the future to improve its applicability and stability in different decision situations.

CRedit authorship contribution statement

Yizhao Xu: Writing – original draft, Methodology, Data curation.
Shuli Yan: Writing – review & editing, Visualization, Supervision.
Feixia Ji: Methodology, Conceptualization.

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Data availability

No data was used for the research described in the article.

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